MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, VOLTAGE COMPENSATED, MULTIFUNCTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers general requirements for a hermetically sealed, 1-1/2 inch diameter, multifunction Temperature Indicator, which is of the Electrical Resistance, Voltage Compensated type, operating on 28 volts D.C. (See 6.1).

1.2 Classification - The indicator shall be furnished in the following ranges:

MS PART NO.	RANGE °C	INDICATOR
MS90322-1	-70 to +150	unlighted
MS90322-2	-70 to +50	unlighted
MS90322-4	-50 to +75	unlighted
MS90322-5	-50 to +300	unlighted
MS90322-8	-70 to +150	lighted
MS90322-9	-70 to +50	lighted
MS90322-11	-50 to +75	lighted
MS90322-12	-50 to +300	lighted

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

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SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical
DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors and Other Uses
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemicals, Films, and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-I-7070	Indicator, Temperature, Electrical Resistance, Hermetically Sealed, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for

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SPECIFICATIONS

Military (Cont'd)

MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection-Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-P-23408	Plating: Tin Cadmium (Electrodeposited)
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-L-27160	Lighting, Instrument, Integral, White, General Specifications for
MIL-C-81013	Cushioning, Material, Transparent, Flexible, Cellular, Plastic Film, for Packaging Applications

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-200	Electron Tube, Selection and Use of
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment

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STANDARDS

Military (Cont'd)

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (For Systems and Equipments)
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS24367	Lamp, Incandescent - Miniature Integral Lighting
MS24515	Lamp, Subminiature
MS28105	Window, Dial - Aircraft Instrument Cover, Glass
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar
MS33639	Case, Instrument, Clamp-Mounted, Aircraft
MS90322	Indicator, Temperature, Electrical Resistance, Voltage Compensated, Multi-Function

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The indicator furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.5).

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3.2 General - The requirements specified in MIL-I-7070 are applicable as requirements of this specification. Additional requirements shall be as specified herein and in accordance with MS90322. In the event the requirements of MIL-I-7070 and this specification conflict, the requirements of this specification shall govern.

3.3 Selection of Government documents - Except as provided in 3.3.1 and 3.3.2, specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3.1 Commercial parts - Commercial parts having suitable properties shall be used, where on the date of invitations for bids, there are not suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, cotter pins, having suitable properties, shall be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

3.3.2 Standard parts - With the exception of 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.4.1 Critical materials - Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose. Noncritical materials shall be used where practicable.

3.4.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.4.3 Metals - Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion by fuels, salt spray, or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

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3.4.3.1 Dissimilar metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless protection against electrolytic corrosion is provided.

3.4.3.2 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.4.3.3 Aluminum alloy parts - Unless otherwise specified, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.4.3.4 Iron and steel parts - Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with QQ-P-416. Parts which are not in sealed cases shall be chromium, nickel, or zinc plated in accordance with QQ-C-320 or QQ-N-290, or QQ-Z-325 respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated in accordance with MIL-P-23408. The class and type of plating shall meet the requirements of 3.4.4.

3.4.4 Protective treatment - When materials are used in the construction of the indicator that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.4.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.6 Fumes and vapors - Materials used in the construction of the indicator shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.5 Design and construction - The indicator shall conform to MS90322. The indicator shall be designed to indicate temperature when used in conjunction with an electrical-resistance circuit. The indicator shall be so constructed that no parts will work loose in service and so that it will withstand the normal shocks, vibrations, and such other conditions as are incident to service without failure during shipping, storage, installation, and operation.

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3.5.1 Maintenance - The design shall be such as to facilitate as much as possible the disassembly, repair or overhaul, service maintenance, and reassembly of the indicator. Tools and items of maintenance equipment shall be made available as commercial standards.

3.5.2 Case - The case shall conform to MS33639 for a nominal diameter of 1-1/2 inches. The case and mounting flange material shall be uniform in texture, having a smooth surface, and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall have sufficient durability to withstand the usage encountered in service.

3.5.2.1 Hermetic sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed, replaced, and the case re-sealed at least three (3) times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any environment to which the indicator may be subjected in normal use in military aircraft (see 6.4.2).

3.5.2.2 Filling medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with BB-N-411, Type I, Class I, Grade C. The filling medium shall contain not more than 0.006 milligram of water vapor per liter (dew point -65° C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 ± 0.1 atmosphere.

3.5.3 Weight - The indicator shall be so designed that when completely assembled the weight shall not exceed 1.0 pound.

3.5.4 Screw threads - Screw threads 0.060 inch in diameter or larger shall be in accordance with MIL-S-7742.

3.6 Electrical requirements -

3.6.1 Primary input power requirements - The indicator shall operate in accordance with MIL-STD-704, Category B, DC power source. The wattage shall not exceed 14.25 watts when operated as specified.

3.6.2 Undervoltage protection - The indicator shall not be damaged by voltages below the minimum specified, and shall automatically resume normal operation when the voltage is within the specified limits.

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3.6.3 Overload protection - Overload protection shall be provided in the indicator. All parts and circuits which are likely to carry an overload as a result of failures or maladjustments shall have suitable protective devices or preferably, shall be proportioned to withstand such overload without damage to the indicator. The indicator shall resume normal operation when the voltage returns within specified limits (after the protective device is replaced).

3.6.4 Electronic parts - Electronic parts comprising the indicator and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists in accordance with MIL-E-5400 shall not be used unless approved by the procuring activity.

3.6.5 Electronic devices - Electronic tubes, transistors and diodes shall be chosen and applied in accordance with MIL-STD-200.

3.6.6 Integral lighting - The lighted indicators shall be provided with 5V integral lighting in accordance with MIL-L-25467 or MIL-L-27160, whichever is specified in the contract or purchase order.

3.6.7 Lamps - The lamps for the lighted indicators shall be located within the indicator case. Lamps conforming to MS24367 or MS24515 may be used in lieu of, or in addition to, the lamps specified in MIL-L-25467 or MIL-L-27160, whichever is specified in the contract or purchase order.

3.6.8 Electric connectors - The electric connector used shall conform to MIL-C-26482 and MS90322. The connector code number markings as specified in MIL-E-5400 shall be legibly and permanently applied to the case or connector of the indicator.

The connector shall be wired as follows:

Terminal	Connection
A	Positive 28V battery
B	Resistance bulb
C	Remaining resistance bulb connection
D	Negative 28V battery
E	5V lighting
F	When required

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3.6.9 Wiring - Internal wiring shall be color coded, neat, and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering, where practicable. The wiring shall be insulated from the instrument case.

3.6.10 Electromagnetic interference - The indicator shall be designed to minimize electromagnetic interference in accordance with MIL-STD-461.

3.6.11 Dielectric strength - The indicator shall be capable of withstanding the application of 200 VDC between the connector and case for a period of 5 seconds.

3.7 Dial and pointer - The dial shall be securely mounted in such a manner that it will not loosen or slip when the indicator is vibrated. If screws are used, they shall be so located that they will not interfere with the proper location of any dial marking. The dial blank shall consist of a flat disc of plastic, metal, or other material which will permit compliance with the lighting and other requirements of this specification.

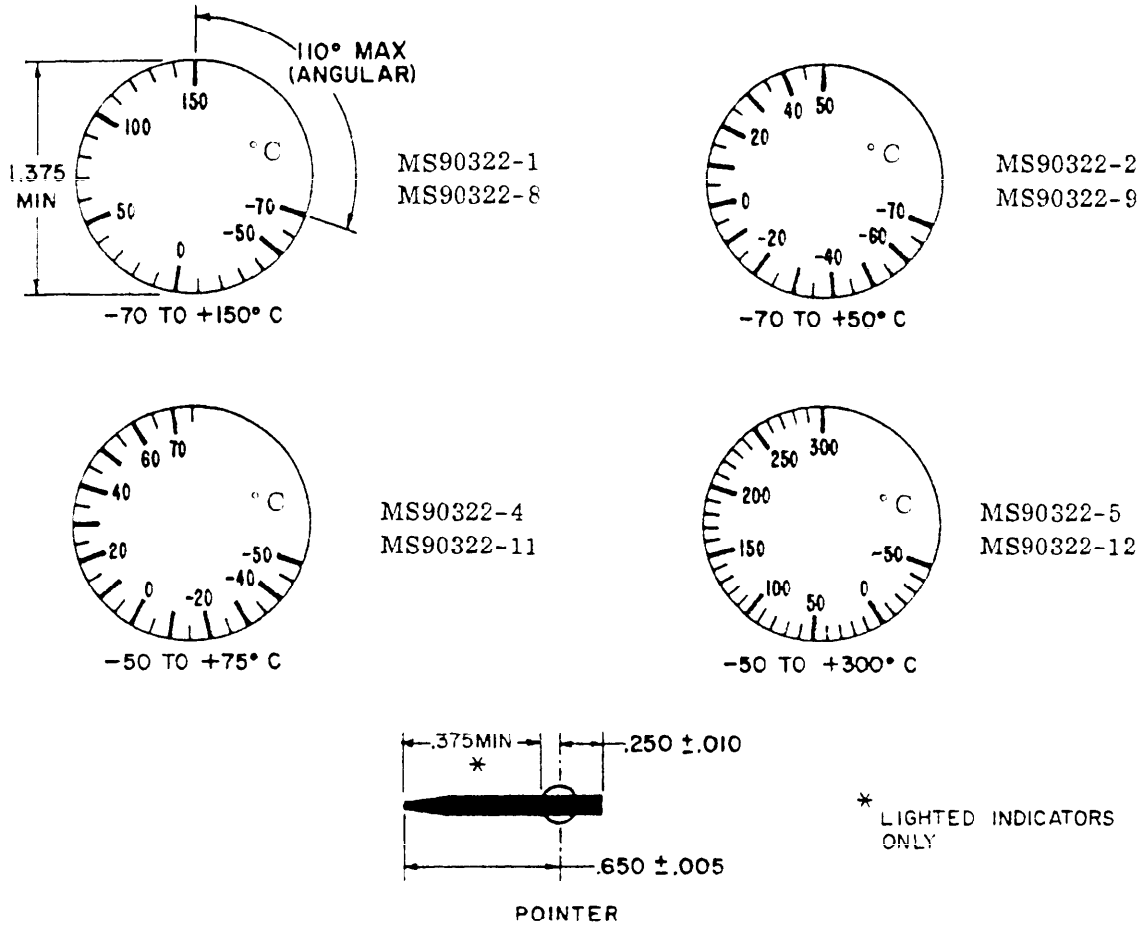
3.7.1 Dial marking - The dial shall be marked as shown in Figure 1. Unless otherwise specified, all visible portions of the dial shall be lusterless black Color No. 37038 of FED-STD-595, and all markings on the dial shall be lusterless white Color No. 37875 of FED-STD-595. All markings shall be sufficiently durable to withstand usage encountered in service. The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be placed so that the center of area of the numeral(s) is on the radial line joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to the graduation to which the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64 inch.

3.7.1.1 Dial visibility - The pointer, numerals, at least 1/32 inch of the shortest graduations, and all other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.7.2 Pointer - The pointer of the indicator shall be as light weight as practicable, sufficiently rigid to prevent oscillation under vibration, and shall be firmly attached to the mechanism. The pointer shall not become damaged or distorted when subjected to the tests specified herein. The shaded portion of the

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horizontal and vertical coordinates of impact, and right



MARKING	HEIGHT OR LENGTH ±0.005	WIDTH OF LINE OR GRADUATION ±0.005	MATERIAL OR FINISH
ALL NUMERALS MAJOR GRADUATIONS MINOR GRADUATIONS LETTERING "°C" (SHADED PORTION OF POINTER)	0.093 0.125 0.093 0.093 ---	0.015 0.025 0.015 0.015 ---	LUSTERLESS WHITE FED-STD-595 COLOR NO. 37875
BACKGROUND OF DIAL (UNSHADED PORTION OF POINTER)	---	---	LUSTERLESS BLACK FED-STD-595 COLOR NO. 37038

DIMENSIONS IN INCHES

FIGURE 1. DIALS AND POINTERS

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pointer shall be finished in lusterless white, Color No. 37875 of FED-STD-595. The unshaded portion shall be finished in lusterless black, Color No. 37038 of FED-STD-595. The pointer shall be suitable for integral lighting and shall conform to Figure 1.

3.7.2.1 Unlighted indicators - The pointer for unlighted indicators shall have a tip of 0.003 inch to 0.005 inch radius. The width of the pointer, 1/4 inch ± 1/64 inch from the tip, shall be 1/8 inch ± 1/64 inch.

3.7.2.2 Lighted indicators - The pointer for lighted indicators shall have a tip of 0.003 inch to 0.005 inch radius. The width of the pointer, 7/16 inch ± 1/64 inch from the tip, shall be 3/32 inch + 1/64 inch. The length of the pointer shall be such that the minimum length of the pointer visible shall be 0.375 inch.

3.7.2.3 Pointer stops - Pointer stops, if used, shall be located to permit the pointer to overswing not less than 3 percent of full scale deflection beyond each end of the scale.

3.8 Cover glass - The quality of the cover glass shall be in accordance with DD-G-451, Type II, Quality AA. Any flaws permitted by DD-G-451 shall not interfere with reading of the dial. The dimensions of the cover glass shall be in accordance with MS28105.

3.8.1 Cover glass mounting - The distance from the inner surface of the cover glass to the surface of the dial on which the marking is applied shall not exceed 0.188 inch. for lighted indicators, and shall not exceed 0.125 inch for nonlighted indicators.

3.8.2 Cover glass coating - The cover glass shall be coated with a reflection reducing film in accordance with MIL-C-14806.

3.9 Performance - The indicator shall perform satisfactorily when subjected to the tests in Section 4 of this specification.

3.9.1 Reliability program - The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.9.1.1 The indicators shall have a mean time between failures (MTBF) of 1000 hours at a 90 percent confidence level.

3.10 Identification of product -

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3.10.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130. A field shall be provided for the FSN but the FSN shall be omitted on Navy procurements.

3.10.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.11 Part numbering of interchangeable parts - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.12 Soldering - Soldering shall be performed in accordance with MIL-S-6872.

3.13 Design data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents referenced in Section 2 unless specified in the contract or purchase order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure the supplies and services conform to prescribed requirements.

4.2 Classification of inspection - Inspection of the indicators shall be classified as follows:

(a) Qualification inspection: Qualification inspection consists of examinations and tests performed on sample indicators submitted for approval as a qualified product.

(b) Quality conformance inspection: Quality conformance inspection consists of examination and tests performed on indicators manufactured and submitted for acceptance under contract.

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4.3 Qualification inspection - The qualification inspection of the indicator shall consist of all the examinations and tests of this specification performed in the order specified under paragraph 4.6 Test Methods and samples submitted for qualification shall be accompanied by the supplier's report showing that the reliability test for the Reliability Qualification Phase para. 4.4.3.1 has been satisfactorily met on other samples of the same equipment.

4.3.1 Qualification inspection samples - Qualification inspection samples shall consist of three (3) indicators manufactured in accordance with this specification. The indicators submitted for qualification inspection shall have been previously subjected only to the individual inspections. The samples shall be forwarded, at the supplier's expense, to the laboratory designated in the letter of authorization. (See 6.5.)

4.3.1.1 Qualification inspection sample identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Qualification Inspection
INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, VOLTAGE COMPENSATED, MULTIFUNCTION
Submitted by (Manufacturer's name, date)
for Qualification Inspection in accordance with MIL-I-81049A under authorization (reference letter authorizing tests).

4.4 Quality conformance inspection - The contractor shall furnish all samples and shall be responsible for accomplishing all the inspections. Sampling Plan B inspections shall be conducted at a Government laboratory designated by the procuring activity. Quality conformance inspection, except for Sampling Plan B, shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality conformance inspection shall consist of the following tests:

(a) Individual inspection

(b) Sampling Plan A inspection

(c) Sampling Plan B inspection

(d) Reliability assurance tests

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4.4.1 Individual inspection - Each indicator submitted for acceptance shall be subjected to Individual inspection. This inspection shall determine compliance with the requirements of material, workmanship and operational adequacy. As a minimum, each indicator accepted shall have passed the following inspections:

Examination of product.	4.6.2
Scale error at room temperature.	4.6.3
Position error.	4.6.4
Friction error.	4.6.5
Hermetic seal.	4.6.6
Insulation leakage.	4.6.7
Integral lighting (lighted indicators only).	4.6.8

4.4.2 Sampling Plans - The Sampling Plans shall consist of Sampling Plan A and Sampling Plan B inspections. The inspection samples selected for sampling tests shall first have passed the Individual inspections. The inspection samples which have been subjected to Sampling Plan A inspection shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual inspections. Inspection samples which have been subjected to the Sampling Plan B inspection shall not be delivered on contract.

4.4.2.1 Sampling Plan A sample selection - Sampling Plan A samples shall be selected at random in accordance with the following schedule:

Quantity Offered for Acceptance	Quantity to be Selected for Inspection
First 15	(See Note)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

When a defective indicator occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected.

NOTE: When Sampling Plan B is invoked, the quantity shall be zero. When Sampling Plan B is to be omitted, the quantity shall be one.

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4.4.2.1.1 Sampling Plan A inspection - Each sample selected for Sampling Plan A inspection shall be subjected to the following tests:

Reduced voltage operation.	4.6.9
Scale error at low temperature.	4.6.10
Scale error at high temperature.	4.6.11
Magnetic effect.	4.6.13
Resistance bulb current.	4.6.15
Vibration resonance.	4.6.16
Light test MIL-L-25467 or MIL-L-27160 (lighted indicators only).	
Open and short circuit.	4.6.26

4.4.2.2 Sampling Plan B instructions - Two (2) indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. Unless otherwise specified in paragraph 6.2, these samples shall be forwarded, at the contractor's expense, to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached and marked with the following information:

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, VOLTAGE COMPENSATED, MULTIFUNCTION
Submitted by (Manufacturer's name, date) for Production Acceptance Sampling Plan B inspection of MIL-I-81049A in accordance with Contract No. _____
Manufacturer's part number _____

4.4.2.2.1 Sampling Plan B inspection - Each sample selected for Sampling Plan B inspection shall be subjected to the following tests:

Sampling Plan A.	
Vibration failure.	4.6.17
Response time.	4.6.18
Dampling factor.	4.6.19
Power consumption.	4.6.20
Bulb lead resistance effect.	4.6.21
Low temperature exposure.	4.6.22
High temperature exposure.	4.6.23
Thermal shock.	4.6.24
Fogging.	4.6.25
Salt spray	4.6.27
Lamp life (lighted indicators only)	4.6.28
Test for helium.	4.6.29
Internal examination.	4.6.30
Lighting contrast (lighted indicators only).	4.6.31

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4.4.3 Reliability assurance tests - Reliability assurance tests shall be conducted using MIL-STD-781. Qualification phase and sampling phase tests shall be conducted. Equipments selected for reliability assurance tests shall first have passed the individual inspections..

4.4.3.1 Reliability qualification phase - Prior to Qualified Products Listing (see 4.3) between three (3) and six (6) indicators shall be tested as outlined in MIL-STD-781 under the section entitled "Qualification Phase of Production Reliability Tests" using Test Plan I.

4.4.3.2 Reliability sampling phase tests - The indicator samples shall be tested as outlined in MIL-STD-781 under the section entitled "Sampling Phase of Production Reliability Tests" using Test Plan I.

4.4.3.3 Reliability test program - At least 30 days prior to the scheduled date of initiation of reliability testing, the contractor shall submit a detailed reliability test program to the procuring activity for approval. Testing shall begin only after formal approval of the test program by the procuring activity.

4.4.3.4 Reliability test samples - A minimum of three (3) sample indicators shall be tested by the contractor for reliability. Duration of the test and accept-reject criteria shall be as specified in MIL-STD-781. Data recording and reporting procedures shall be as specified in MIL-STD-781.

4.4.3.5 Reliability demonstration test procedure - The test equipment and fixtures shall include provisions for vibration and temperature cycling. The direction of the vibration is not critical. The vibration double amplitude shall be .010 to .015 inch with a frequency of 45 to 55 cycles per second to produce a vibration level of 1.5 g $\pm$ 10 percent. The indicators shall be subjected to operational cycles which consist of 4 hours of operation with power and input resistance applied and 30 minutes with power off. The operational cycles shall be conducted in temperatures of +25 $\pm$ 5° C, +55 $\pm$ 5° C, and -55 $\pm$ 5° C in the following sequence: 4 hours at -55° C, 30 minutes off; 4 hours at +25° C, 30 minutes off; 4 hours at +55° C, 30 minutes off; 4 hours at +25° C, 30 minutes off, etc. During the operating portion of the cycle the input resistance shall be varied to produce pointer cycling consisting of deflections from -40 $\pm$ 5° C to 50 $\pm$ 5 percent of full scale and return to -40 $\pm$ 5° C at a rate of 2.5 $\pm$ 0.5 cycles per minute. The indication and operational cycles shall be repeated throughout the test until an accept or reject decision is reached. The continuity of the test may be interrupted for weekend shutdown. During each 4 hour operation cycle the indicators shall be vibrated for a minimum period of 10 minutes each hour.

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4.4.3.6 Performance characteristics to be measured - Each indicator shall be subjected to the scale error test at room temperature once each 48 hours of operation during the +25° C temperature cycle. The scale error recorded throughout the test shall not exceed the tolerances specified in Table I during reliability testing.

4.4.3.7 Failure criteria - All failures which prevent compliance with any requirements of this specification shall be counted as relevant failures.

4.4.3.8 Preventive maintenance - During the period of the tests no preventive maintenance measures may be performed upon the indicator.

4.4.3.9 Disposition of samples upon completion of tests - Any sample tested may be delivered on contract provided it meets all the following requirements:

(a) It is representative of production units currently being accepted.

(b) It is in "good as new" condition or has been refurbished. (See 6.4.3.)

(c) It is otherwise satisfactory.

4.5 Test Conditions -

4.5.1 Standard test conditions - Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient +25 ± 5° C
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient up to 90 percent relative humidity
Input Power	28.5 ± 0.5 VDC

4.5.2 Test readings - Unless otherwise specified, before a test reading is taken, the indicator shall be vibrated using a vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.

4.5.3 Test tolerances - The tolerances specified in Table I, column A, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table I, column B, are applicable only when the indicators are subjected to a specific test independent of sequence.

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4.5.4 Attitude - Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.5.5 Integral lighting - The lighting system of the lighted indicator shall be energized during all tests except Salt spray or as otherwise specified herein and shall operate satisfactorily when energized. The lighting system shall operate satisfactorily after completion of each test unless otherwise specified herein.

4.6 Test Methods -

4.6.1 Individual test methods - The individual test methods shall be conducted in accordance with MIL-I-7070 unless otherwise specified herein.

4.6.2 Examination of product - Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.3 Scale error at room temperature - The test resistance specified in Table I, for the temperature range being tested, shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied, and the scale error shall not exceed the tolerances specified in Table I, for the temperature range being tested.

4.6.4 Position error - When the indicator is subjected to the position error test, the maximum change in indication shall not exceed the tolerances specified in Table II, for the temperature range being tested. With no voltage applied to the indicator, the pointer should be off scale at the low temperature end.

4.6.5 Friction error - The friction error test may be conducted in conjunction with the scale error test. The friction error of the indicator shall not exceed the tolerances specified in Table II, for the temperature range being tested.

4.6.6 Hermetic seal - The indicator shall be immersed in a suitable liquid, such as water. The absolute pressure of the air above the liquid shall then be reduced to approximately 1 inch Hg and maintained for 1 minute, or until air bubbles substantially cease to be given off by the liquid, whichever is longer. The absolute pressure shall then be increased to 2 1/2 inch Hg. Any bubbles coming from within the indicator case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air on the various parts of

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TABLE I

TEST TOLERANCES

Indicator temperature range	Test point ° C	Test resistance (ohms)	Scale error during reliability testing (tolerances) ° C ±	Scale error at room temp. (tolerances) ° C ±		Reduced voltage (tolerances) ° C ±		Scale error at low temp. (tolerances) ° C ±		Scale error at high temp. (tolerances) ° C ±		Scale error at room temp. after open & short circuit (tolerances) ° C ±		Scale error at room temp. after vibration cycling (tolerances) ° C ±		Scale error with bulb lead resistance (tolerances) ° C ±		Extreme low temp. exposure				Scale error at room temp. after high temp. exposure (tolerances) ° C ±	
																		Scale error at low temp. (tolerances) ° C ±		Scale error at room temp. after low temp. exposure (tolerances) ° C ±			
				A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B
-70 to +150	-70	68.27	6	3	3	2	2	7	7	7	7	10	4	10	7	6	6	14	7	10	3	10	3
	-30	80.56	6	3	3	2	2	6	6	6	6	10	4	10	7	6	6	13	6	10	3	10	3
	0	90.38	4	2	2	2	2	4	4	4	4	10	3	10	6	6	6	12	4	10	2	10	2
	30	100.91	6	3	3	2	2	6	6	6	6	10	4	10	7	6	6	13	6	10	3	10	3
	80	120.36	6	3	3	2	2	6	6	6	6	10	4	10	7	6	6	13	6	10	3	10	3
	120	137.78	6	3	3	3	3	6	6	6	6	10	4	10	7	6	6	13	6	10	3	10	3
	150	151.91	6	3	3	3	3	7	7	7	7	10	4	10	7	6	6	14	7	10	3	10	3
-70 to + 50	-70	68.27	4	2	2	2.5	2.5	3.5	3.5	3.5	3.5	3.5	3	2.5	2.5	3.5	3.5	5	3.5	3.5	2	3.5	2
	-50	74.24	4	1.5	1.5	.5	.5	3	3	3	3	3.0	2.5	2.0	2.0	3	3	4.5	3	3.0	1.5	3.0	1.5
	-30	80.56	2	1.0	1.0	.5	.5	2	2	2	2	2.5	2.0	1.5	1.5	3	3	3.5	2	2.5	1.0	2.5	1.0
	-10	87.04	2	1.0	1.0	.5	.5	2	2	2	2	2.5	2.0	1.5	1.5	3	3	3.5	2	2.5	1.0	2.5	1.0
	0	90.38	2	1.0	1.0	.5	.5	2	2	2	2	2.5	2.0	1.5	1.5	3	3	3.5	2	2.5	1.0	2.5	1.0
	+10	93.80	2	1.0	1.0	1.0	1.0	2	2	2	2	2.5	2.0	1.5	1.5	3	3	3.5	2	2.5	1.0	2.5	1.0
	30	100.91	3	1.5	1.5	1.0	1.0	3	3	3	3	3.0	2.5	2.0	2.0	3	3	4.5	3	3.0	1.5	3.0	1.5
	50	108.39	3	2	2	2.5	2.5	3.5	3.5	3.5	3.5	3.5	3	2.5	2.5	3.5	3.5	5	3.5	3.5	2	3.5	2
-50 to +75	-50	74.24	3	1.5	1.5	1	1	3	3	3	3	2	2	4.0	3.5	3	3	5.5	3	4.0	1.5	4.0	1.5
	-30	80.56	3	1.5	1.5	1	1	3	3	3	3	2	2	4.0	3.5	3	3	5.5	3	4.0	1.5	4.0	1.5
	-10	87.04	3	1.5	1.5	1	1	3	3	3	3	2	2	4.0	3.5	3	3	5.5	3	4.0	1.5	4.0	1.5
	0	90.38	2	1.0	1.0	1	1	2	2	2	2	1.5	1.5	3.5	3.0	3	3	4.5	2	3.5	1.0	3.5	1.0
	+10	93.80	3	1.5	1.5	1	1	3	3	3	3	2	2	4.0	3.5	3	3	5.5	3	4.0	1.5	4.0	1.5
	30	100.91	3	1.5	1.5	2	2	3	3	3	3	2	2	4.0	3.5	3	3	5.5	3	4.0	1.5	4.0	1.5
	50	108.39	3	1.5	1.5	2	2	3	3	3	3	2	2	4.0	3.5	3	3	5.5	3	4.0	1.5	4.0	1.5
	75	118.29	4	2	2	2.5	2.5	3.5	3.5	3.5	3.5	2.5	2.5	4.5	4.0	3.5	3.5	6.0	3.5	4.5	2.0	4.5	2.0
-50 to +300	-50	37.95	10	5	5	4	4	10	10	10	10	7	7	12	10	10	10	15	10	12	5	12	5
	0	50.00	10	5	5	4	4	10	10	10	10	7	7	12	10	10	10	15	10	12	5	12	5
	+50	63.55	10	5	5	4	4	10	10	10	10	7	7	12	10	10	10	15	10	12	5	12	5
	100	79.00	10	5	5	4	4	10	10	10	10	7	7	12	10	10	10	15	10	12	5	12	5
	150	96.50	8	4	4	4	4	7	7	7	7	7	7	12	10	10	8	15	7	12	4	12	4
	200	116.50	10	5	5	5	5	10	10	10	10	7	7	12	10	10	10	15	10	12	5	12	5
	250	139.00	10	5	5	5	5	10	10	10	10	7	7	12	10	10	10	15	10	12	5	12	5
	300	164.50	10	5	5	5	5	10	10	10	10	7	7	12	10	10	10	15	10	12	5	12	5

Column A - Tolerances applicable only when indicators are subjected to tests in a specified sequence.
Column B - Tolerances applicable only when indicators are subjected to a specific test independent of sequence.

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the case shall not be considered as a leak. A helium leak detector or other means of test equal or superior in sensitivity to the immersion test method described above, may be used upon approval by the procuring activity. Where a leak detector is employed for conducting this test, the leak rate of the indicator shall not exceed 1 micron-cubic foot per hour at a pressure differential of approximately 1 atmosphere.

4.6.7 Insulation leakage - The insulation resistance shall be equal to or greater than that specified in Table II, for the temperature range being tested.

4.6.8 Integral lighting - Each indicator shall be subjected to and shall meet the requirements of the following tests specified in MIL-I-25467 or MIL-L-27160 whichever is applicable:

(a) Operation.

(b) Dial visibility (see para. 3.7.1.1 of MIL-I-81049A for viewing angle).

4.6.9 Reduced voltage operation - The indicator shall be subjected to a scale error test at room temperature with an applied voltage of 22.5VDC at the test points specified in Table I. The indications at the reduced voltage shall not differ from the indications with 28.5VDC applied by more than the tolerances specified in Table I, for the temperature range being tested.

4.6.10 Scale error at low temperature - The scale error shall not exceed the tolerances specified in Table I, for the temperature range being tested.

4.6.11 Scale error at high temperature - The scale error shall not exceed the tolerances specified in Table I, for the temperature range being tested.

4.6.12 Dielectric strength - The applied potential shall be 200 VDC between the connector and case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test.

4.6.13 Magnetic effect - The magnetic compass shall not deflect more than that specified in Table II, for the temperature range being tested.

4.6.14 Electromagnetic interference - The electrical circuitry of the indicator shall be tested for conducted and radiated noise for interference in accordance with MIL-STD-461.

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TABLE II

TEST TOLERANCES

Indicator temperature range	Position error (° C)	Friction error (° C)	Insulation leakage (megohms)	Compass deflection (° ang)	Resistance bulb current (ma)	Vibration resonance		Response time (sec.)	Damping factor	Power consumption (watts)	Insulation resistance after salt spray (megohms)
						Pointer deviation (° C)	Pointer oscillation (° C)				
-70 to +150	3	3	20	5	22	2	2	3	1.6	1.5	20
-70 to +50	1.5	1.5	20	5	22	1.5	1	3	1.6	1.5	20
-50 to +75	2.5	2.5	20	5	22	2.5	2.5	3	1.6	1.5	20
-50 to +300	7	7	20	5	22	7	7	3	1.6	1.5	20

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4.6.15 Resistance bulb current - A total resistance of 100 ohms shall be applied across the resistance bulb terminals of the indicator, the test voltage applied, and the resistance bulb circuit current measured. The current shall not exceed the current given in Table II, for the temperature range being tested.

4.6.16 Vibration resonance - The vibration resonance test shall be in accordance with Procedure IV of MIL-E-5272. Resistance and voltage shall be applied to the indicator to produce a reading. The indicator shall be subjected to vibration with a total amplitude between 0.009 and 0.011 of an inch. The pointer oscillation and deviation shall not exceed the tolerances specified in Table II, for the temperature range being tested.

4.6.17 Vibration failure - The vibration failure test shall be in accordance with Procedure V of MIL-E-5272. Resistance and voltage shall be applied to the indicator to produce a reading. After the vibration test, the indicator shall be subjected to a scale error test at room temperature. The scale error at room temperature after a vibration shall not exceed the tolerances specified in Table I, for the temperature range being tested.

4.6.18 Response time - The response time shall not exceed the tolerances specified in Table II, for the temperature range being tested.

4.6.19 Damping factor - The damping factor shall not be less than that specified in Table II, for the temperature range being tested.

4.6.20 Power consumption - The power consumption shall not exceed tolerances specified in Table II, for the temperature range being tested.

4.6.21 Bulb lead resistance effect - The indicator shall be connected as shown on Figure 2 and subjected to a scale error test at room temperature. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance by more than the tolerances specified in Table I, for the temperature range being tested.

4.6.22 Low temperature exposure - The scale error at low temperature shall not exceed the tolerances specified in Table I. The scale error at room temperature after low temperature exposure shall not exceed the tolerances specified in Table I, for the temperature range being tested.

4.6.23 High temperature exposure - The scale error at room temperature after high temperature exposure shall not exceed the tolerances specified in Table I, for the temperature range being tested.

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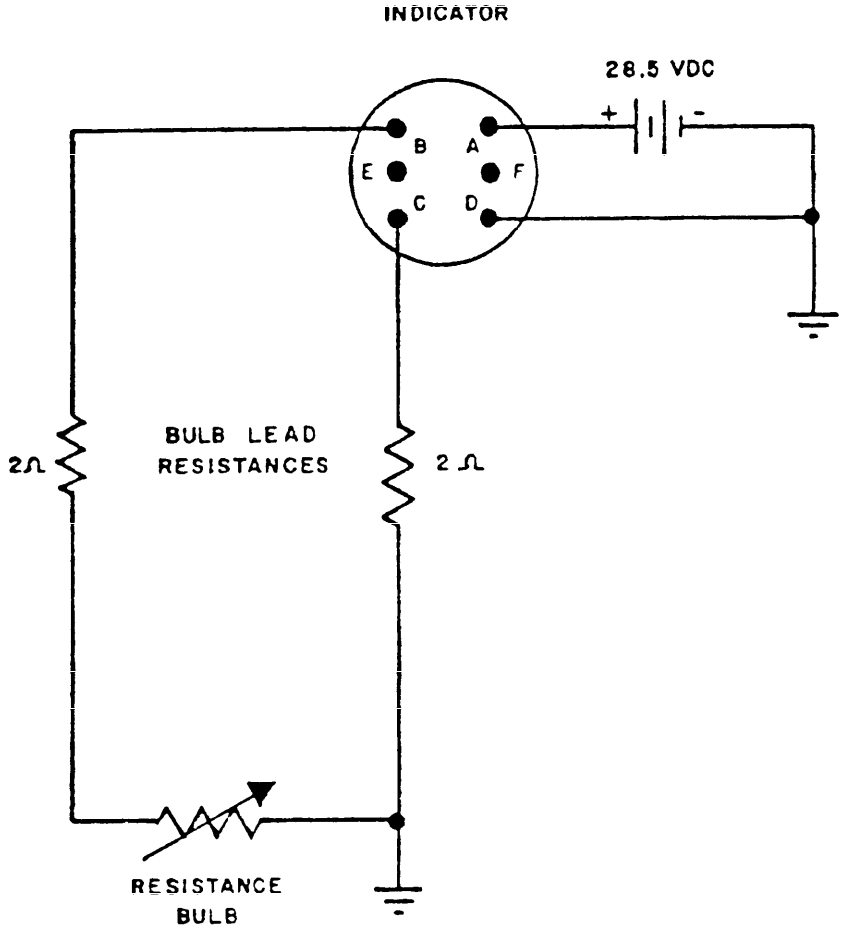


FIGURE 2. WIRING DIAGRAM FOR BULB LEAD RESISTANCE TEST

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4.6.24 Thermal shock - The thermal shock test shall be in accordance with MIL-I-7070 except that the Hermetic seal test shall be as specified herein.

4.6.25 Fogging - The indicator shall be energized and placed in a 71 ± 2°C controlled ambient temperature for a minimum period of one (1) hour. After this period, and while still at this temperature, an ice cube shall be rubbed on the cover glass for a period of one to two minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog inside the cover glass. Evidence of fogging shall be cause for rejection.

4.6.26 Open and short circuit - The scale error after open and short circuit test shall not exceed the tolerances specified in Table I, for the temperature range being tested.

4.6.27 Salt spray - The indicator shall be subjected to a salt spray test. Following salt spray exposure, the indicator shall be resubjected to and shall meet the provisions of Insulation leakage and Hermetic seal tests of this specification.

4.6.28 Lamp life - The lighting circuit of the indicator shall be subjected to the Life test as specified in MIL-L-25467 or MIL-L-27160 whichever is applicable.

4.6.29 Test for helium - The indicator case shall be punctured, or the filling tube cut, and the indicator subjected to the test for helium by means of a mass spectrometer type helium detector. Failure to detect helium shall be cause for rejection.

4.6.30 Internal examination - The case of the indicator shall be opened and the internal mechanism examined. The indicator shall also be examined in accordance with the Lamp Circuit Test as specified in MIL-L-25467 or MIL-L-27160 whichever is applicable. Any deterioration or damage which could in any manner prevent the indicator from meeting functional operation or maintenance requirements during service life shall be cause for rejection. The indicator shall be resealed if no damage is found.

4.6.31 Lighting contrast - The indicator shall be subjected to and meet the requirements of the Contrast Test specified in MIL-L-25467 or MIL-L-27160 whichever is applicable.

4.7 Rejection and retest - Indicators which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and action taken to correct the defects shall be furnished to the Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - The indicator shall be packaged in accordance with MIL-STD-794, Level A, unless otherwise specified in the contract or purchase order (see 6.2). The method of preservation shall be in accordance with MIL-P-116, Method III.

5.1.1 Level A - Each Indicator shall be preserved and packaged in accordance with MIL-P-116, Method III and shall be cushioned with material conforming to Type II, MIL-E-81013, in sufficient thickness to provide adequate protection within the unit container. The unit container shall conform to Grade 6Ws, PPP-B-636.

5.2 Packing - The indicator shall be packed in accordance with MIL-STD-794, Level A, Load Type 3, unless otherwise specified in the contract or purchase order (see 6.2).

5.2.1 Level A - Each indicator shall be packed in weather-resistant shipping containers conforming to PPP-B-636. The grade of each shipping container shall be determined by the gross weight and size and limited to the special requirements of Table II of the specification. Waterproofing shall be in accordance with paragraph 30.4 of the appendix of PPP-B-636. Where practicable, the exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.3 Marking - The interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking - In addition to any other markings required by the contract or purchase order, the following markings shall appear on two (2) opposite sides of the interior and exterior containers.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The indicator covered by this specification is intended for use on aircraft to measure various fluctuations in temperature.

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6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) The quantity, MS part number and dash number of the indicator desired.

(c) Levels of packaging and packing desired.

(d) Government laboratory or testing activity (see 4.3, 4.4, 4.4.2.2).

(e) When Sampling Plan B tests will not be performed by a Government laboratory (see 4.4.2.2).

6.3 Definitions -

6.3.1 Amplitude - Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.3.2 Hermetic seal - A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.3 Good as new - "Good as new" shall mean indicators operated less than 10 percent of the specified MTBF operation.

6.3.4 Refurbished - "Refurbished" shall mean that the indicator has been completely overhauled with all component parts meeting current parts standard; and the indicator shall have been subjected to and met all the requirements of a new indicator.

6.4 Qualification - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids; qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible

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for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity.

6.5 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 Asterisks - Asterisks are not used in this revision to identify the extensive changes with respect to the previous issue.

Custodians:
Army - EL
Navy - AS
AF - 11

Review activities:
Army - AV, MU

Preparing activity:
Navy - AS

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE	
MIL-I-81049A	INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, VOLTAGE COMPENSATED, MULTIFUNCTION

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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